

Section 2: System Design Fundamentals

Domain Name System (DNS)

DNS or Domain Name System, is one of the fundamental building blocks of the internet is domain name systems. In this section we will be exploring how the DNS works, how domain names get translated into IP addresses, how caching improves performance, and why DNS plays a crucial role in large-scale systems.

By the end of this session, you will not only be able to understand the DNS resolution process, but also gain insights into the optimization and security considerations. It's an essential topic for both system design and technical interviews.

What is DNS?

DNS is often called the phone book of the internet. Just like we store phone numbers under contact names in our phones, DNS helps us access websites using human-readable domain names instead of IP addresses.

For example: When you type google.com in our browser, your computer doesn't actually understand google.com, what it does is, it needs an IP address, something like 142.240.190.78, to locate the google server. DNS is the one that makes the translation actually happen behind the scenes.



Without DNS, we will have to memorize a long string of IP addresses for every website. Imagine remembering different IP addresses for all these websites that we are accessing on a daily basis. That will be a nightmare for us. DNS also helps to scale the internet efficiently also. The internet consists of billions of devices. The DNS ensures that they communicate smoothly by organizing domain names into structured distributed systems.

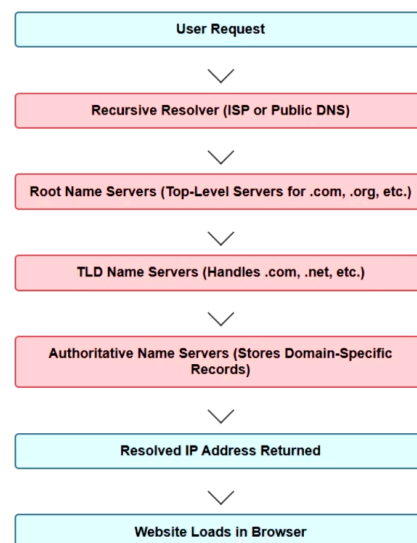
In short, DNS is the foundation that keeps the internet user-friendly and scalable.

Types of DNS servers.

There are many types of DNS server used today. If you look at it, when we talk about DNS , there has to be some server that is actually keeping track of these mappings. For example, this name google.com has to resolve to this IP address, this name amazon.com has to resolve to this IP address.

So essentially we have DNS servers, but there are multiple types of servers. So let's first understand what all types of servers are there.

- **Root Name Server:** These are the top level servers that handle all domain extensions like .com, .org, .net, etc. They don't store the actual domain records, but it directs all the requests to appropriate second level servers, which are called TLD name servers.
- **TLD Name Server:** These are the second-level servers, which is a set of servers where whenever the request reaches a TLD server, it redirects the query to an authoritative name server. So the TLD server keeps track of, lets say we have a TLD server for .com, search query for google.com reaches this, then TLD name server's responsibility is to figure out which actual server has this mapping of google.com to the IP address, which is the third level called authoritative name servers.
- **Authoritative Name Server:** These are the third level servers also called as authoritative name servers. These are the actual servers that store the DNS records like A record, CNAME record, MX record, where they have the mapping for what domain and IP address is there. Once a request reaches the authoritative server, it returns the correct IP address to the user.
- **Recursive Resolvers:** These are the final level of servers which are called recursive resolvers. These resolvers actually handle the DNS queries on behalf of the user. So typically ISPs set up recursive resolvers at their end and predominantly their responsibility is to forward the request to the respective authoritative name server and also to cache the domain names for whatever requests have been resolved already, which will provide a little bit of efficiency.



So, whenever the user makes a request, we will first check with the recursive resolver server whether it is there or not. If not, then we will go to the top and figure out from where this domain name can be resolved and then the actual IP address will be returned to the user.

DNS Caching & Performance Optimization

DNS caching helps in performance optimization. Imagine if every time you wanted to call a friend, you had to pick up their number from the directory. That would be very slow and inefficient. Instead, if you just save their number in your phone, you just have to dial that contact information.

DNS caching somehow works in the same way. It helps to reduce latency. Instead of performing the full DNS lookup every time, the cache records allow the devices to resolve domain names almost instantly. It also reduces the load on the DNS servers. Fewer queries means less traffic to the DNS servers, which will improve the overall performance of the internet.

DNS caching happens at multiple levels. First, once you make the request, the request goes all the way to the domain server and it gets resolved and you open that site in your browser. So the browser also caches the DNS records. The next time you make a request for the same domain, it will first look at the browser caching and cache it from there. The operating system is also caching the DNS records. So there is one more level of caching. And then finally, there are these recursive resolver cache servers. The ISPs have the DNS resolver caches. It keeps all the resolved domains in its cache and it serves all the next requests for all those domains from that cache.

One important thing that we need to understand when we are talking about DNS caching is TTL (Time-To-Live). It is basically the expiration timer for the cache record. So every DNS record has a TTL value, which determines how long the cache is valid before it expires.

A short TTL ensures that all the cache records have up to date IP addresses for every domain. On the contrary, a long one will actually reduce the queries to the DNS server, making it more optimal, but any update in the domain to IP will be delayed.

In summary, DNS caching significantly improves performance and efficiency. However, managing TTL values is the key to balancing speed and accuracy. There has to be some balance between the performance and the accuracy of your records.

The Domain Name Resolution Process

Let's say you type google.com in your browser and hit enter. First your browser checks its own cache. If you recently visited google, your browser may already have the IP address stored. So it can skip the lookup and load the page instantly. If your browser cache doesn't have it, maybe the operating system cache is there. So the computer also keeps a local cache of DNS records, which reduces the need for external lookups.

Now let's say the OS also doesn't have it, then a query is sent to the local DNS resolver, usually provided by your internet service providers. This is the first step in reaching out to the broader DNS network, which is outside of your home network.

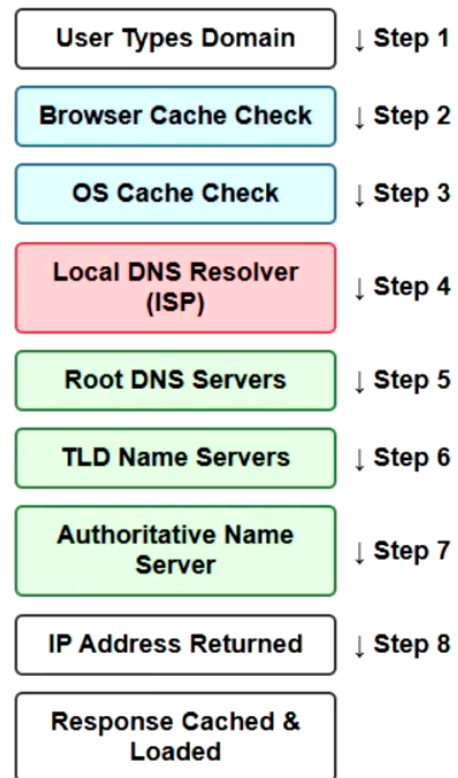
If the local DNS server doesn't have the required domain cached, it reaches out to the root DNS servers. These are the backbone of the DNS system, redirecting the request to appropriate top-level domain servers. The top level domain servers like .com, .org or .net respond with the details of the authoritative DNS server, which has the actual record, the actual domain to IP mapping.

Then the authoritative DNS server is hit, it will find what IP is to be sent back for the requested DNS. It contains the actual IP address for the google.com, so it returns to the resolver.

Then the local DNS resolver caches it at their ISPs and sends it back to your OS and your browser will cache it, and your browser will then connect to that IP, and the website will load.

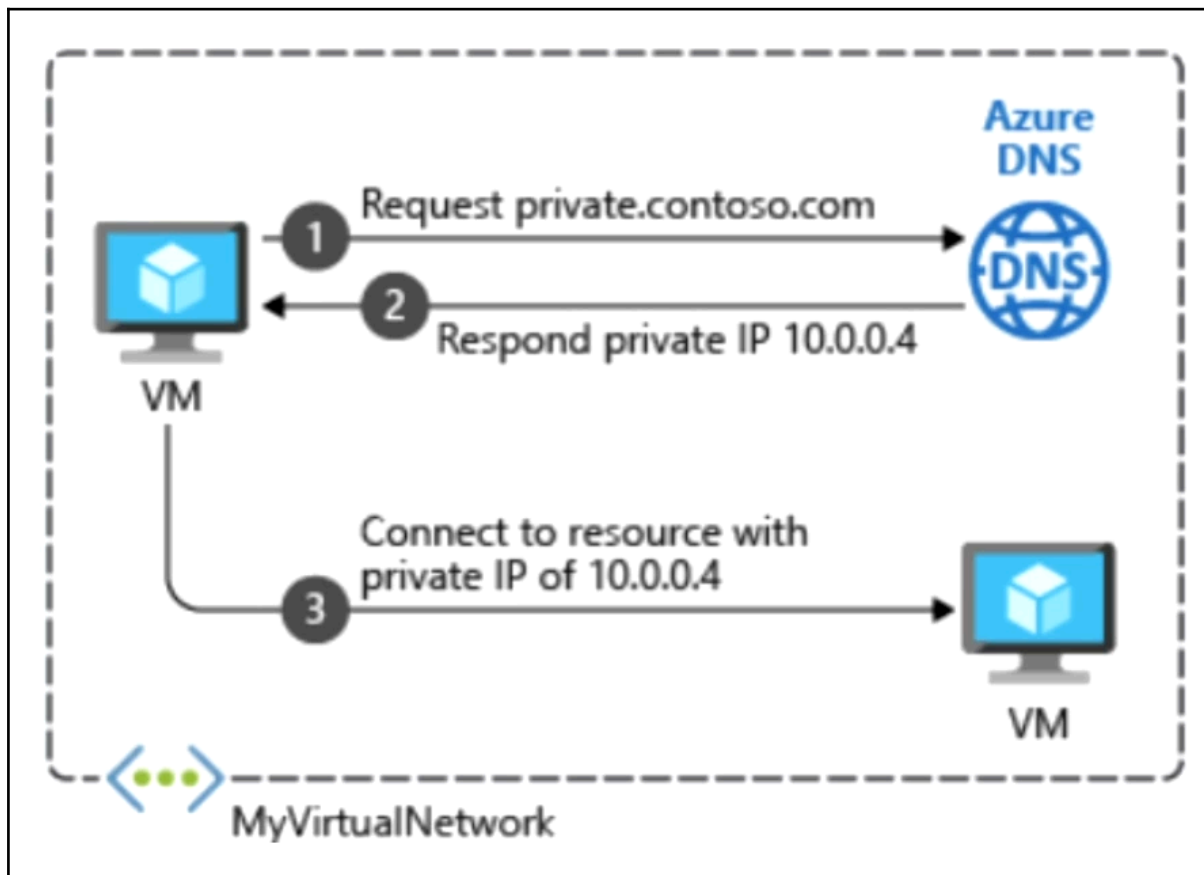
All this happens in milliseconds, and thanks to DNS caching the next time we are visiting, this process will be even faster. So that's the complete process of domain name resolution.

Step-by-Step DNS Resolution



Importance of DNS Large-Scale Systems

In small scale applications, DNS simply resolves domain names to IP addresses. But in large-scale systems, DNS plays a much bigger role in ensuring reliability, performance and security. For global applications like google, amazon, netflix, downtime is not an option.

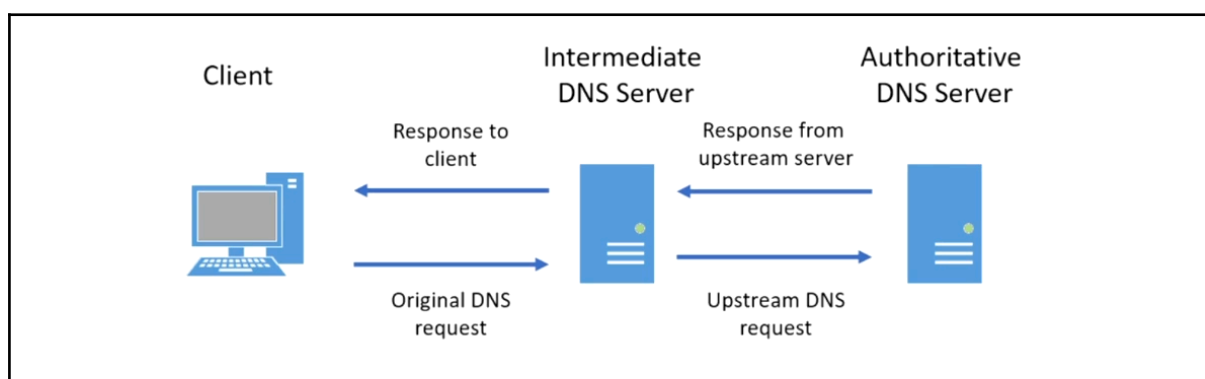


- **Ensuring High Availability:** DNS-based load balancing is the first benefit that we can get. Instead of pointing all the traffic to a single server, DNS can actually distribute requests across multiple servers worldwide, preventing the overloading of any single server and ensuring smooth performance. Second is Anycast DNS. Instead of a single DNS server responding to queries, multiple geographically distributed servers can handle the DNS resolution request. This ensures that the users get a response from the nearest server, which will be the fastest server for them, and it will reduce latency.
- **DNS Failover Strategies:** There are actually 2 types/categories of DNS servers, let's say primary and secondary DNS servers. If the primary DNS server fails, requests are automatically routed to backup, which will ensure the continuity of your domain name resolution functionality. There are health checks and failovers configured there. Some DNS providers monitor their server health and automatically redirect traffic to the healthy server. If the primary is not healthy, if it goes down, it will actually send it to the secondary server automatically.

- **Content Delivery Network (CDNs) & DNS:** DNS helps content delivery networks or CDNs to optimize their performance. When you visit a website using a CDN, the DNS will redirect you to the nearest server instead of a central one. For example, if a user in India accesses a website hosted in New York, a CDN can serve the content from a local data center in Mumbai, which will reduce the latency and improve load time. As for how this happens, let's say my CDN is called `www.example.com`. So when my domain name resolution happens, the DNS will make sure to resolve it to the nearest server to load my content, nearest in terms of geographically, improving the overall performance.
- **DNS Security Risks:** DNS is also a target for cyber attacks. So DNS poisoning and cache poisoning attackers, there is a possibility for attackers to inject fake DNS records to the malicious site. So instead of the actual IP address, they can actually replace it with the malicious site's IP address, and the moment you type the domain, they can take you to the malicious website. DDoS attacks on the DNS servers are very common. Attackers flood DNS servers with requests, making websites unreachable. So if you type a domain name and the DNS server is under DDoS attack and you are not able to resolve the IP address of that particular website, you will never be able to reach that website. There are some secure DNS protocols like DNSSEC, which will help us verify the authenticity and prevent the tampering of DNS records.

In summary, DNS is more than just a lookup service. It's a key player in performance, scalability and security.

Summary & Takeaways



- DNS is critical for internet functionality.
- Caching improves efficiency and reduces latency.
- DNS load balancing & failover are essential for large-scale systems.
- Security threats exist, but best practices help mitigate them.
- What's Next?
 - Client-Server Model

Interview Questions Related to DNS

- *Explain the DNS resolution process step by step.*
- *What is the difference between authoritative and recursive DNS servers?*
- *How does DNS caching improve performance? Where does it occur?*
- *What is TTL in DNS, and why is it important?*
- *How does DNS-based load balancing work in large-scale systems?*
- *What are the common DNS-related security threats, and how can they be mitigated?*